An Evidence-Hardening Funnel for Transcriptomic Prognostic Claims in Clear Cell Renal Cell Carcinoma

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Abstract

Public transcriptomic studies often treat large tumor–normal expression changes as candidate prognostic biomarkers without testing whether the associations are externally reproducible or driven by tissue composition. We developed HYDRA-ccRCC, a reproducible evidence-hardening workflow for clear cell renal cell carcinoma. TCGA-KIRC RNA sequencing was used for discovery; SVA-aware paired analyses of GSE40435 and GSE53757 tested tumor–normal replication; and GSE29609 and E-MTAB-1980 provided survival checks. Following external statistical review, proportional-hazards tests were retained as diagnostics rather than exclusion gates, and 1,000 patient bootstraps estimated adjusted Cox coefficient uncertainty without repeated significance selection. The revised analysis identified 3,304 reproducible differentially expressed genes, 536 strict candidates, and 27 high-confidence candidates. A separate apeglm sensitivity removed the hard fold-change gate and globally corrected survival tests across all 32,916 count-QC genes without redefining the primary set. E-MTAB-1980 mapped 26 candidates, 23 retained the TCGA hazard direction, and 13 had same-direction FDR support in both unadjusted and limited-adjustment models; GSE29609 was not broadly supportive. All 27 bootstrap intervals excluded zero, and all 27 candidates retained direction and FDR support after published consensus tumor-purity adjustment. A TRACERx Renal sensitivity linked 186 primary-tumor regions from 73 patients. Regional high/low discordance affected 32.8–51.7% of multiregion tumors, while reducing the resampled cohort from 73 to 39 patients lowered the median direction-agreement fraction from 99.3% to 86.8%. A randomized CheckMate 025 analysis linked all candidates to 250 patients, but no expression-by-treatment interaction survived FDR correction for overall or progression-free survival. ACADM and CRYL1 showed the cleanest metabolic prognostic convergence, while DDC remained high-priority but cohort-dependent. HYDRA-ccRCC prioritizes candidate prognostic associations while exposing cohort disagreement, composition dependence, spatial sampling sensitivity, and the absence of corrected treatment-effect modification; it does not establish a validated biomarker panel.

Keywords clear cell renal cell carcinoma, TCGA, GEO, transcriptomics, survival analysis, bootstrap uncertainty, external evaluation

1 Introduction

Clear cell renal cell carcinoma (ccRCC) is shaped by VHL loss, hypoxia-inducible signaling, angiogenesis, altered metabolism, and a heterogeneous tumor microenvironment [14, 4]. These processes produce broad transcriptomic differences between tumor and normal kidney. A large differential-

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expression effect, however, can reflect tumor identity, loss of normal nephron compartments, immune or vascular admixture, or technical context rather than stable prognostic information.

HYDRA-ccRCC tests a narrow question: among genes reproducibly dysregulated across ccRCC expression cohorts, which have survival associations that remain credible after clinical adjustment, diagnostic proportional-hazards assessment, coefficient-uncertainty estimation, external outcome evaluation, and cell-source triangulation? The contribution is an auditable evidence funnel rather than a new prediction algorithm or a fixed multigene panel.

2 Methods

2.1 Cohorts and Revised Analysis Design

TCGA-KIRC public, de-identified data were obtained from the Genomic Data Commons through TCGAbiolinks [5]. Discovery used 541 primary tumors and 72 solid-tissue normal samples. GSE40435 (101 tumor and 101 normal samples) and GSE53757 (72 tumor and 72 normal samples) were obtained from the Gene Expression Omnibus [2] for tumor–normal replication.

GSE29609, with 39 tumors and 17 deaths, was retained as a small exploratory survival-direction check. E-MTAB-1980, with 101 patients and 23 deaths, provided the larger external evaluation. The candidate rule was revised in response to statistical review, not in response to gene-specific external outcomes, but earlier project versions had already inspected both cohorts. We therefore describe the current set as a reviewer-driven reanalysis rather than a prospectively frozen or blinded validation set.

2.2 Differential Expression and Reproducibility

TCGA raw unstranded STAR counts were analyzed with DESeq2 [10]. Genes required at least 10 counts in at least 10 samples. Significance required Benjamini–Hochberg adjusted $p < 0.05$ and absolute $\log_2$ fold change of at least 1. To assess instability from noisy maximum-likelihood fold changes and hard inclusion boundaries, a separate sensitivity analysis estimated MAP $\log_2$ fold changes with apeglm [15], applied no absolute fold-change cutoff, fit the fully adjusted Cox model to every count-QC gene represented in the variance-stabilized matrix, and controlled FDR once across that complete survival-test universe. This sensitivity result did not redefine the primary candidate set.

The GEO cohorts were analyzed with limma [12]. GSE40435 used parsed patient-pair identifiers; GSE53757 used the documented alternating tumor–normal sample order. For each cohort, surrogate-variable analysis (SVA) [9] protected the tumor–normal contrast with a full patient-plus-condition model and a null patient model; estimated surrogate variables were added to the limma design. Probe expression was collapsed to gene symbols by mean expression. A reproducible DEG required TCGA significance, the same tumor–normal direction in both GEO cohorts, and nominal $p < 0.05$ in at least one GEO cohort.

2.3 TCGA Survival Selection

Overall survival models used TCGA primary tumors and standardized continuous expression. The main Cox model [6] adjusted for age, sex, pathologic stage, and collapsed grade among complete cases. Stage-complete and grade-complete sensitivity models were fit separately. High-confidence candidates required main-model FDR < 0.01, absolute log hazard ratio at least $\log(1.5)$, non-trivial GEO effect support, and concordant nominal support in both sensitivity models. Schoenfeld-residual tests were reported as diagnostics rather than exclusion criteria; coefficients with evidence of non-proportionality were interpreted as average hazard effects over follow-up.

2.4 External Survival Testing

External expression values were standardized within cohort. GSE29609 tested unadjusted continuous-expression Cox associations and retained missing and contradictory results. E-MTAB-1980 tested all revised candidates with an unadjusted primary model and a limited secondary model adjusting for age, high T stage, and metastatic status. Candidate-wise multiplicity was controlled with Benjamini–Hochberg FDR. Strict external support required TCGA-concordant direction and FDR support in both external models. Proportional-hazards tests remained visible diagnostics but were not support gates.

2.5 Multiregion Transportability Sensitivity

Processed TRACERx Renal tumor TPM values, region annotations, and clinical data were retrieved from a public repository pinned to commit `3fefd80a` [7]. The analysis retained primary-tumor regions with linked overall-survival data and mapped all high-confidence candidates by gene symbol. Expression was transformed as $\log_2(\mathrm{TPM}+1)$. For each gene, the exploratory high/low threshold was the median of patient-level regional medians, which prevented patients with more sampled regions from determining the cutoff. A tumor was discordant when at least one region fell below the threshold and another fell at or above it.

One thousand repeats selected one primary region per patient and fit an unadjusted Cox model using standardized continuous expression. A second scenario selected event-stratified subsets of 39 patients, the size of GSE29609, before choosing one region per patient. These subsets contained nine deaths because they preserved the TRACERx event fraction. The resulting quantiles describe estimate dispersion under region selection and cohort subsampling; they are not confidence intervals for a fixed population parameter.

2.6 Randomized Treatment-Interaction Analysis

Normalized pretreatment RNA expression and linked clinical data were obtained from the public supplementary workbook of Braun et al. for CheckMate 025, a randomized comparison of nivolumab with everolimus in previously treated advanced ccRCC [3]. Each candidate was standardized across the combined RNA-profiled trial subset. For overall survival and progression-free survival, the randomized model contained treatment, continuous expression, and their interaction. A limited sensitivity model additionally adjusted for age, sex, and MSKCC risk group. Benjamini–Hochberg correction was applied across the 27 interaction tests within each endpoint and model. Arm-specific expression coefficients and proportional-hazards tests were reported as descriptive diagnostics, not as substitutes for the interaction estimate.

2.7 Coefficient Uncertainty and Held-Out Clinical Increment

Coefficient uncertainty used 1,000 event-stratified patient bootstraps. Each repeat refit the age-, sex-, stage-, and grade-adjusted Cox coefficient for every high-confidence candidate. We recorded empirical standard errors, percentile 95% intervals, bias, and direction agreement without applying per-repeat p-value, FDR, PH, or selection-frequency gates.

Predictive increment used 20 repeats of five-fold event-stratified cross-validation. For each candidate, a clinical-only Cox model and a clinical-plus-one-gene model were trained within each fold and evaluated on held-out patients. Out-of-fold predictions were pooled within each repeat, and the concordance difference was summarized across repeats. These repeated-split distributions are internal uncertainty summaries, not independent confidence intervals or evidence of clinical utility.

2.8 Composition and Cell-Source Checks

Clinical-plus-gene models were compared with models that additionally included proximal-tubule, endothelial, immune, and stromal expression-derived marker scores. Candidate-specific leave-one-out scores reduced direct circularity when a candidate overlapped a marker set. These scores are composition screens, not direct tumor-purity estimates.

Published TCGA consensus purity estimates from Aran et al. [1] were matched by aliquot identifier. Each revised candidate was refit with standardized purity, age, sex, stage, and grade. The analysis recorded direction agreement, false-discovery rate, proportional-hazards diagnostics, and attenuation relative to the original clinical model.

Human Protein Atlas v25.1 normal-tissue single-cell expression was mapped to renal epithelial, vascular, immune, stromal, and other compartments using a prespecified rule set. The analysis recorded the dominant mapped compartment and top cell types for every revised candidate. Because the source contains normal tissues rather than ccRCC tumor cells, it supports cell-source triangulation but cannot establish tumor-cell expression or mechanism.

2.9 Reproducibility and Validation

The scripted pipeline records public source URLs, access dates, input checksums, random seeds, package versions, and output checksums. Automated tests check required columns, expected candidate coverage, cohort counts, monotonic funnel behavior, and impossible values. The manuscript is compiled separately from the analysis pipeline.

3 Results

3.1 The Evidence Funnel Compresses Broad Dysregulation

TCGA analysis identified 14,337 significant Ensembl-level genes and 8,852 genes after symbol mapping. GSE40435 and GSE53757 identified 1,076 and 3,139 significant genes, respectively. SVA estimated zero additional surrogate variables in both patient-blocked GEO designs, so their fitted design ranks were unchanged. Cross-cohort filtering retained 3,304 reproducible DEGs. Of these, 1,176 passed main Cox FDR, 1,113 passed sensitivity requirements, 536 met the strict definition, and 27 met the high-confidence definition.

The no-hard-LFC sensitivity successfully modeled all 32,916 count-QC genes. A single Benjamini–Hochberg correction across those tests retained 12,135 genes at survival FDR below 0.05, including 1,188 of the 3,304 primary reproducible DEGs. The large number of adjusted survival associations argues against treating the sensitivity hits as a replacement panel; its role is to show the effect of removing the upstream boundary while using stabilized fold-change estimates.

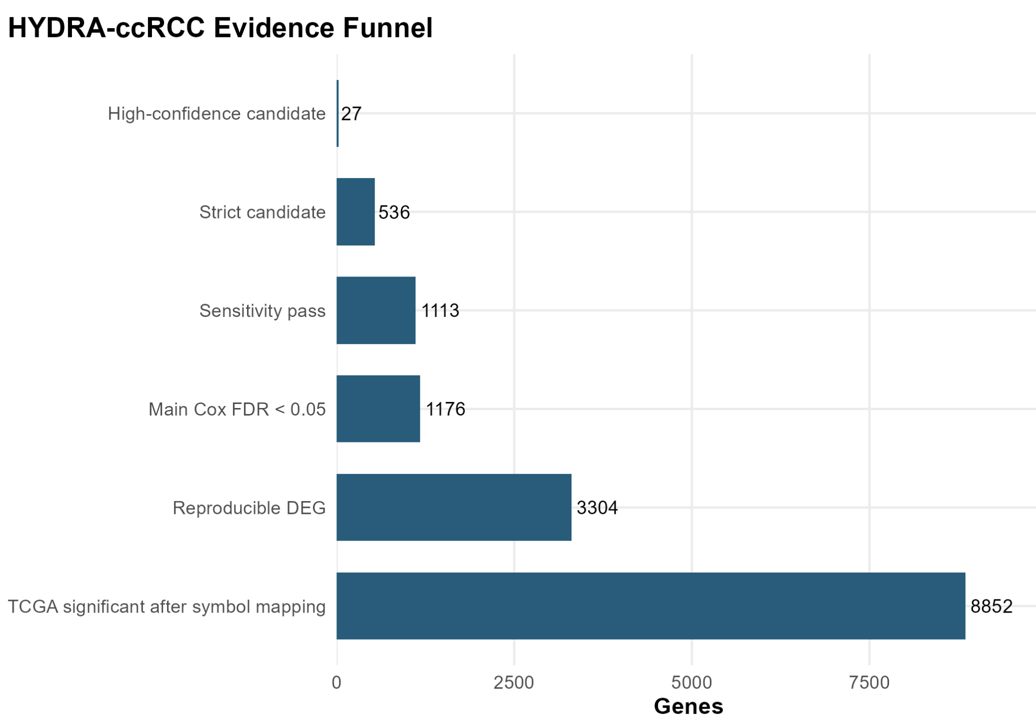

Figure 1: Evidence funnel from TCGA tumor–normal differential expression to the revised high-confidence candidate set. Proportional-hazards diagnostics are intentionally absent as a filtering step.

Differential-expression magnitude and adjusted survival effect were not equivalent. Many protective candidates were lower in tumor, consistent with retained renal epithelial or metabolic differentiation, while several risk-direction genes were compatible with immune, inflammatory, vascular, or stromal states.

Table 1: Discovery evidence funnel

Metric	Count
TCGA tumor / normal samples	541 / 72
TCGA significant mapped genes	8,852
Reproducible DEGs	3,304
Main Cox FDR pass	1,176
Sensitivity pass	1,113
Strict candidates	536
High-confidence candidates	27

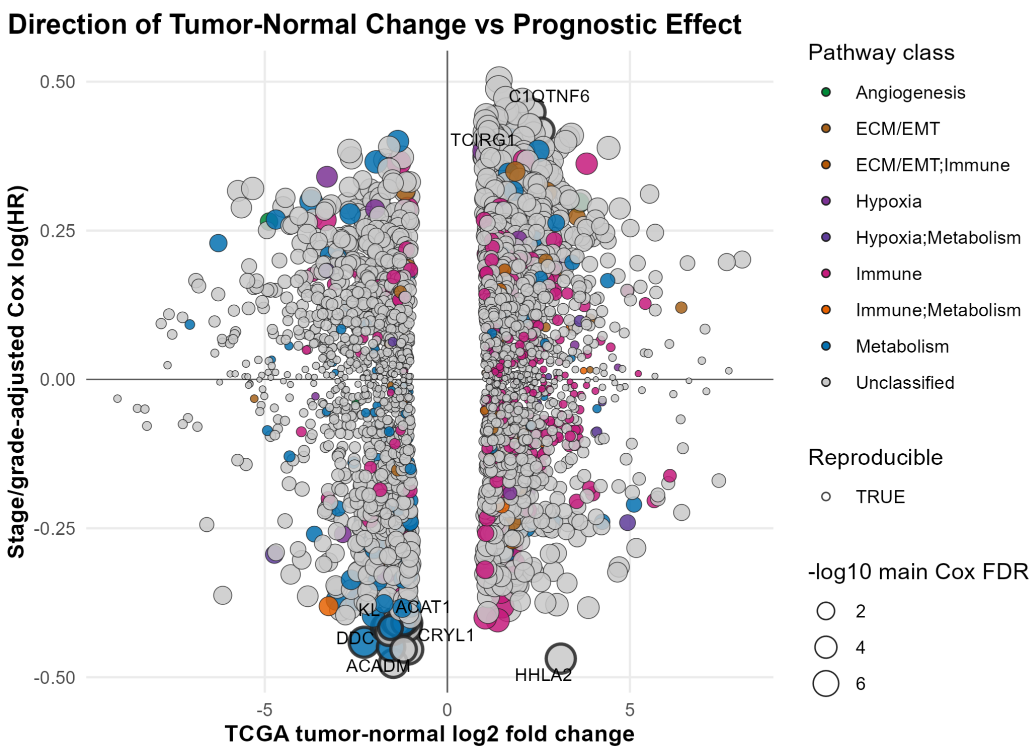

Figure 2: Directional relationship between tumor–normal expression change and adjusted TCGA survival effect.

3.2 Independent Survival Cohorts Give Unequal Support

GSE29609 tested 25 platform-present candidates. Six retained the TCGA hazard direction, none had same-direction nominal support, and DDC, ACAT1, CLCN5, TCIRG1, and RBM47 had nominal opposite-direction associations. Its 17 events and distinct measurement platform limit gene-specific inference, but the result rejects a claim of universal cross-platform validation.

E-MTAB-1980 mapped 26 candidates. Twenty-three retained the TCGA direction, 15 had same-direction unadjusted FDR support, and 13 retained adjusted FDR support and passed the revised external rule. No candidate had a nominally significant opposite-direction association.

The revised strict E-MTAB-1980 set was DDC, CRYL1, ACADM, KL, ACAT1, CLCN5, TCIRG1, GJB1, TEK, EMCN, PODXL, FUT6, and HIBCH. Eight of the 13 had PH diagnostic p-values of at least 0.05 in both external models. KL, ACAT1, and CLCN5 had nominal PH diagnostics in the unadjusted model, while TCIRG1 and GJB1 had nominal diagnostics in both models; these five entered the revised support set because PH diagnostics no longer acted as exclusion gates. The increase from eight to 13 is therefore a change in statistical classification rather than five new replications. Statistical agreement also did not make the genes biologically equivalent: TEK and

Table 2: External survival evidence

Metric	GSE29609	E-MTAB-1980
Samples / events	39 / 17	101 / 23
Candidates mapped	25	26
Same TCGA direction	6	23
Same-direction nominal	0	15
Strict external support	—	13
Opposite-direction nominal	5	0

EMCN were vascular signals, PODXL and GJB1 were compartment-sensitive, and TCIRG1 carried an immune-composition interpretation.

3.3 Multiregion Sampling Widens Candidate Effects

The TRACERx sensitivity included 186 primary regions from 73 patients with 16 deaths; 58 patients had at least two regions. All 27 candidates mapped. Under the exploratory median readout, candidate-level regional discordance ranged from 32.8% to 51.7% of multiregion tumors. ACADM was discordant in 19 of 58 patients (32.8%), CRYL1 in 24 (41.4%), and DDC in 26 (44.8%). These rates quantify cutoff crossing rather than continuous-expression measurement error and depend on the cohort-relative threshold.

In the full-cohort one-region analysis, 26 of 27 candidates retained the TCGA hazard direction at the median; PANK1 was the exception. Across candidates, the median direction-agreement fraction was 99.3%, and central 95% resampling intervals excluded zero for 18 candidates. In 39-patient subsets with nine deaths, the median direction-agreement fraction fell to 86.8%, and only four intervals excluded zero. ACADM retained the TCGA direction in 100.0% of full-cohort repeats and 97.5% of smaller-cohort repeats. CRYL1 fell from 99.5% to 86.8%, while DDC fell from 98.0% to 80.4%; their smaller-cohort intervals crossed zero.

3.4 Randomized Data Do Not Establish Treatment Effect Modification

The CheckMate 025 RNA subset contained 250 patients, including 120 assigned to nivolumab and 130 to everolimus, with 191 overall-survival and 222 progression-free-survival events. The overall nivolumab-versus-everolimus hazard ratio was 0.675 for overall survival (95% CI 0.506–0.899; $p = 0.0073$) and 0.790 for progression-free survival (95% CI 0.605–1.033; $p = 0.085$), confirming that treatment coding recovered the expected overall-survival advantage in this molecular subset.

No candidate-by-treatment interaction survived FDR correction for either endpoint in the randomized or adjusted model. ACADM had a nominal overall-survival interaction in the unadjusted model (interaction HR per SD, 0.746; 95% CI 0.559–0.996; $p = 0.0467$; FDR $= 0.393$), but the adjusted interval crossed one (HR, 0.748; 95% CI 0.554–1.009; $p = 0.0571$; FDR $= 0.385$). CRYL1 and DDC had no nominal interaction for either endpoint. Thus, the randomized comparison distinguishes the candidates' prognostic evidence from unproven treatment-predictive value.

3.5 Bootstrap Intervals Support Coefficient Direction

All 27 candidates completed all 1,000 event-stratified patient bootstraps. Every median bootstrap coefficient retained its full-fit direction, and every percentile 95% interval excluded zero. Empirical standard errors ranged from 0.067 to 0.105. These distributions quantify uncertainty in the adjusted coefficients; they were not converted into another candidate-selection rule.

3.6 Single Genes Add Small Internal Discrimination

Across repeated held-out splits, the clinical-only model had mean concordance 0.745. All 27 candidates produced a positive mean concordance change when added individually; 26 had a positive empirical 2.5th percentile, while FHOD1 crossed zero. The largest mean increments were CYFIP2 (+0.0219), PANK1 (+0.0218), CRYL1 (+0.0206), ACADM (+0.0181), HHLA2 (+0.0176), and

CLCN5 (+0.0175). These small internal gains do not define a multigene model or show clinical usefulness.

3.7 Cell Source Changes the Candidate Hierarchy

All 27 candidates retained TCGA hazard direction after bulk marker-score adjustment, but only 18 remained significant at FDR < 0.05. PODXL, EMCN, FUT6, DDC, KL, HIBCH, GJB1, and ACAT1 had the largest loss of statistical support, suggesting sensitivity to renal epithelial, vascular, or other composition.

Consensus purity estimates matched 516 complete-case TCGA tumors with 170 deaths. After direct purity adjustment, all 27 candidates retained their original hazard direction and remained significant at FDR < 0.05. The largest relative attenuation in absolute log hazard ratio was 8.9%, for GRAMD1A. Measured consensus purity therefore did not explain the TCGA candidate associations.

Human Protein Atlas data mapped all 27 genes. TEK and EMCN were vascular-dominant; CYFIP2, GRAMD1A, TCIRG1, and RBM47 were immune-dominant. LTB4R was renal-epithelial dominant under the prespecified mapping, while GJB1 had renal epithelial expression but a higher maximum in another normal-tissue compartment. These observations support compartment-aware interpretation but do not establish ccRCC tumor-cell origin.

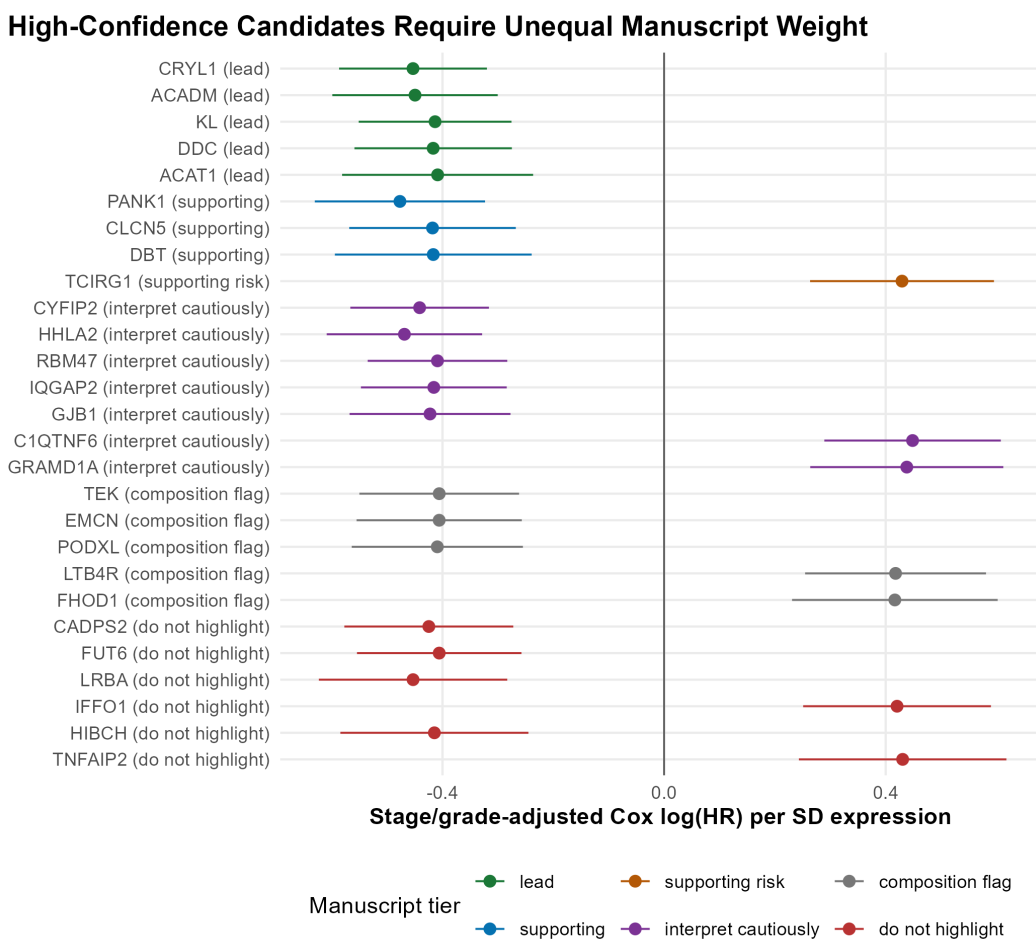

Figure 3: Adjusted TCGA associations for the revised 27-candidate set, grouped by manuscript role.

3.8 Integrated Interpretation Separates Consistent and Conditional Evidence

ACADM and CRYL1 had the cleanest convergence of TCGA association, bootstrap coefficient support, positive held-out increment, strict E-MTAB-1980 support, and no nominal PH diagnostic in either E-MTAB-1980 model. Their direction is consistent with loss or retention of renal epithelial metabolic differentiation. DDC remains a high-priority metabolic hypothesis, but its opposite-direction nominal association in GSE29609 and loss of marker-score-adjusted FDR support make its evidence explicitly cohort- and composition-dependent.

KL, ACAT1, and CLCN5 met the revised external rule only after PH diagnostics ceased to be gates; all three had a nominal PH diagnostic in the unadjusted E-MTAB-1980 model, ACAT1 and CLCN5 were opposite-direction nominal findings in GSE29609, and KL and ACAT1 lost marker-score-adjusted FDR support. Among the three newly retained TCGA candidates, RBM47 was contradicted in GSE29609 and immune-dominant in normal-tissue single-cell data, GJB1 combined strict E-MTAB-1980 support with non-proportionality and composition sensitivity, and LTB4R lacked strict external support. None is elevated to a primary lead solely because the corrected selection rule retained it.

The findings do not support a generic hypoxia-only explanation. VHL/HIF biology can reshape oxidative metabolism and tissue state, but the candidate evidence more directly reflects metabolic differentiation plus immune and vascular composition. The results prioritize experiments and literature review; they do not establish a mechanism or therapeutic target.

4 Discussion

HYDRA-ccRCC shows why prognostic transcriptomic claims need several distinct tests. Expression replication reduced broad TCGA dysregulation, survival selection reduced it further, and external evaluation separated a supported subset from candidates that remained cohort-dependent. The all-gene apeglm sensitivity removes the discontinuity at the primary fold-change boundary, but its broad survival signal does not provide a principled replacement candidate definition. The reviewer-driven revision also changed how uncertainty is used: bootstrapping now estimates adjusted coefficient precision, while cross-validation is reserved for held-out discrimination. Neither procedure creates an additional significance-selection gate.

The two external survival cohorts should be interpreted together. GSE29609 supplied a useful contradictory result, but its small event count and microarray coverage limit power. E-MTAB-1980 provided stronger directional and multiplicity-controlled support, but its 23 deaths still constrain adjustment and precision. Thirteen revised external hits are evidence for candidate-level agreement, not evidence for a deployable panel.

The TRACERx results make spatial sampling a measurable qualification rather than a generic concern. High/low classifications changed across regions for at least 32.8% of multiregion tumors, yet hazard direction was usually stable when all 73 patients were retained. The 39-patient scenario widened the effect distributions and weakened direction agreement, especially for CRYL1 and DDC, while ACADM remained comparatively stable. This pattern supports the narrower interpretation proposed from multiregion ccRCC work: regional sampling noise becomes more consequential as cohort information decreases [8, 7]. It does not identify sampling as the cause of GSE29609 disagreement because the cohorts also differ in platform, case mix, events, and follow-up.

ACADM and CRYL1 are the most consistent next-stage metabolic focus. DDC remains worth studying because its strong TCGA and E-MTAB-1980 results and bootstrap interval coexist with explicit contradiction in GSE29609 and composition sensitivity, making it a conditional rather than uniformly replicated lead. The five genes added to the E-MTAB-1980 support set by removal of PH gating should likewise be interpreted through their time-averaged effects, cross-cohort results, and cell-source evidence rather than promoted on the revised flag alone.

The CheckMate 025 analysis directly separates prognosis from treatment effect modification. The nominal ACADM overall-survival interaction is biologically interesting because higher ACADM expression was associated with lower hazard in the nivolumab arm but not the everolimus arm. Its FDR near 0.39, loss of nominal significance after limited adjustment, and null progression-free-survival interaction make it hypothesis-generating only. CRYL1 and DDC show no randomized evidence of differential benefit. RCC still provides a precedent for the distinction: sarcomatoid differentiation is associated with poor prognosis, yet a post hoc CheckMate 214 analysis found substantially better outcomes with nivolumab plus ipilimumab than with sunitinib in this subgroup [13, 11]. A future confirmatory analysis should prespecify a small biomarker set or metabolic score and an interaction model in an independent randomized immunotherapy cohort.

5 Limitations

This retrospective study uses public bulk transcriptomic data. TCGA supplies discovery, selection, the all-gene boundary sensitivity, bootstrap coefficient estimation, and repeated internal prediction assessment. The global all-gene Cox result reuses TCGA for expression and survival analysis and is sensitivity evidence rather than external validation. The bootstrap intervals are conditional on the revised full-data candidate set and do not account for uncertainty in upstream candidate selection. E-MTAB-1980 has only 23 deaths and permits limited covariate adjustment; GSE29609 has 17 deaths, different technology, and incomplete candidate coverage. Neither external cohort has multiregion sampling. TRACERx provides that structure but contributes only 73 survival-linked patients and 16 deaths, so its Cox models are unadjusted and exploratory. Its median threshold is cohort-relative, and its 39-patient subsets contain nine rather than 17 deaths; the analysis therefore cannot recreate GSE29609 or apportion its disagreement among sampling, platform, and clinical differences. The external cohorts had been inspected before this reviewer-driven reanalysis, so their results cannot be described as blind prospective validation. The CheckMate 025 interaction analysis is post hoc, covers an RNA-profiled trial subset rather than every randomized patient, uses archived normalized expression, and tests 27 candidates across two endpoints. Two patients lacked age for adjusted models. No corrected interaction is confirmatory, and the everolimus comparator does not address current first-line immunotherapy combinations.

Adjacent normal kidney is not equivalent to healthy kidney. Consensus purity is inferred from several molecular and pathology estimates and is not direct histopathologic measurement in this analysis. HPA v25.1 contains normal-tissue rather than ccRCC tumor single-cell data. Bulk RNA cannot separate tumor-cell-intrinsic expression from nephron retention, immune infiltration, endothelial content, stromal admixture, or necrosis. RNA abundance also does not demonstrate protein effect, causation, mechanism, or therapeutic relevance.

6 Conclusion

The revised workflow narrows thousands of reproducible tumor–normal changes to 27 high-confidence candidates, evaluates them in two external survival cohorts, quantifies adjusted-coefficient uncertainty with patient bootstrapping, estimates held-out clinical increment with cross-validation, adjusts directly for consensus tumor purity, triangulates normal-tissue cell source, measures multi-region sampling sensitivity, and tests treatment effect modification in randomized CheckMate 025 data. ACADM provides the most stable integrated metabolic prognostic evidence and a nominal treatment-interaction hypothesis, but no candidate has multiplicity-controlled evidence of differential nivolumab benefit. CRYL1 remains a strong but small-cohort-sensitive prognostic lead, while DDC remains conditional. The appropriate output is a prioritized set of candidate prognostic associations and a transparent record of where the evidence remains fragile, not a validated biomarker panel.

Data Availability

All source data are public and identified by TCGA-KIRC, GSE40435, GSE53757, GSE29609, E-MTAB-1980, Human Protein Atlas v25.1, TRACERx Renal, CheckMate 025 supplementary data, and the Aran et al. TCGA purity supplement. TRACERx files are pinned to the repository commit recorded in `../results/tables/tracerx_multiregion_source_files.csv`; the CheckMate workbook checksum is recorded in `../results/tables/checkmate025_source_file.csv`. Download URLs, access dates, and checksums are recorded in `../results/tables/source_provenance.csv`; generated tables and figures are stored under `../results/`; and the one-command entry point is `../run_pipeline.ps1`.

Ethics Declaration

This secondary analysis used public, de-identified datasets and involved no new recruitment, intervention, or access to identifiable private information. Final competition or publication requirements should be confirmed with the relevant review body.

Author Contributions

The student researcher is responsible for the project question, methodological decisions, verification of the analysis, interpretation, and final writing. Contributions must be reviewed and updated before submission to match the student's documented work.

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Conflict of Interest

The author reports no conflict of interest.

AI Assistance Disclosure

OpenAI Codex assisted with code auditing, implementation support, debugging, reproducibility checks, and revision of this internal development manuscript. The student remains responsible for understanding and independently verifying the code, statistical decisions, citations, and scientific claims. AI-generated prose must not be copied directly into IRIS or ISEF submission materials.

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